

**CSES Problem Set****Salary Queries**TASK | [SUBMIT](#) | [RESULTS](#) | [STATISTICS](#) | [TESTS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

A company has  $n$  employees with certain salaries. Your task is to keep track of the salaries and process queries.

**Input**

The first input line contains two integers  $n$  and  $q$ : the number of employees and queries. The employees are numbered  $1, 2, \dots, n$ .

The next line has  $n$  integers  $p_1, p_2, \dots, p_n$ : each employee's salary.

After this, there are  $q$  lines describing the queries. Each line has one of the following forms:

- $! k x$ : change the salary of employee  $k$  to  $x$
- $? a b$ : count the number of employees whose salary is between  $a \dots b$

**Output**

Print the answer to each  $?$  query.

**Constraints**

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq p_i \leq 10^9$
- $1 \leq k \leq n$
- $1 \leq x \leq 10^9$
- $1 \leq a \leq b \leq 10^9$

**Example**

Input:

```
5 3
3 7 2 2 5
```

**Range Queries**

...

[Forest Queries](#)

-

[Hotel Queries](#)

-

[List Removals](#)

-

[Salary Queries](#)

-

[Prefix Sum Queries](#)

-

[Pizzeria Queries](#)

-

[Visible Buildings Queries](#)

-

[Range Interval Queries](#)

-

...

**Your submissions**

? 2 3

! 3 6

? 2 3

Output:

3

2